The current state of single-cell proteomics data analysis

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Computational toolbox

scpdata

15 curated datasets from published work ready to be analyzed in R

scp data structure

single cell + proteomics

Bioconductor

scp

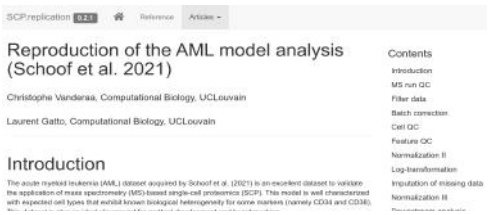
Functionality to process and analyze mass spectrometry-based single-cell proteomics data

- > Consistent input and output
- > High-quality documentation
- > Exemplary coding practices

Python integration

In practice:

SCP.replication



Website with articles that replicate published analyses using scpdata + scp

- > 8 articles with documented code
- > Label-free and multiplexed data
- > DDA and DIA data

uclouvain-cbio.github.io/SCP.replication

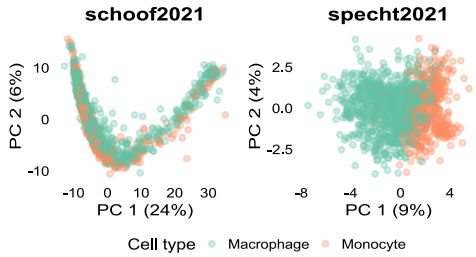
lgatto.github.io/QFeaturesScpWorkshop2021/

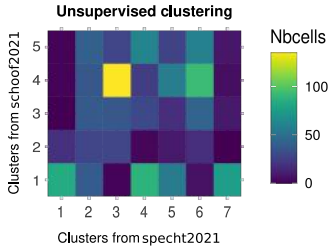
youtu.be/CLZxjIKikIU

Workflow example

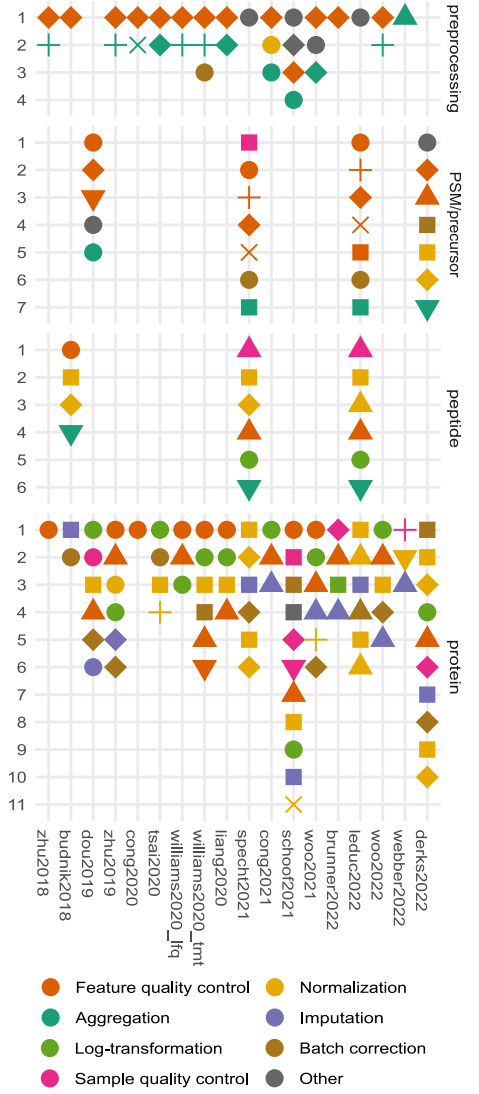
```r
## Load data
library(scpdata)
scp <- schoof2021()

library(scp)
## Feature QC
scp <- filterFeatures(scp, ~ !isContaminant)

## Sample QC
failed <- c("F117", "F44", "F5", "F58")
scp <- subsetByColData(scp, !scp$File.ID %in%
                                        failed)

## Batch correction (+ quantitative thresholding)
library(reticulate)
library(zellkonverter)
spt <- import("sceptre")
sce <- getWithColData(scp, "proteins")
adata <- SCE2AnnData(sce, X_name = 1)
spt$normalize(adata)
sce <- AnnData2SCE(adata)
scp <- addAssay(scp, sce, name = "proteins_norm")
scp <- addAssayLink(scp, from = "proteins",
                    to = "proteins_norm",
                    varFrom = "Accession",
                    varTo = "Accession")

## Sample QC
library(scuttle)
qc <- perCellQCMetrics(scp[["proteins_norm"]],
                       assay.type = 1)
scp$outlier <- isOutlier(qc$sum, nmads = 2,
                         log = TRUE, type = "both")
scp$detected <- qc$detected
scp <- subsetByColData(scp, !scp$outlier &
                               scp$detected > 700)

## Normalization
sf <- colSums(assay(scp, "proteins_norm"))
sf <- sf / median(sf)
scp <- sweep(scp, MARGIN = 2, STATS = sf, FUN = "/",
             i = "proteins_norm",
             name = "proteins_medshift")

## Log-transformation
scp <- logTransform(scp, base = 2, pc = 1,
                    i = "proteins_medshift",
                    name = "proteins_log")

## Imputation
sce <- getWithColData(scp, "proteins_log")
adata <- SCE2AnnData(sce, X_name = 1)
spt$impute(adata)
sce <- AnnData2SCE(adata)
scp <- addAssay(scp, sce, name = "proteins_imput")
scp <- addAssayLink(scp, from = "proteins_log",
                    to = "proteins_imp",
                    varFrom = "Accession",
                    varTo = "Accession")

## Normalization
means <- rowMeans(assay(scp, "proteins_imp"))
scp <- sweep(scp, MARGIN = 1, STATS = means,
             FUN = "-", i = "proteins_imput",
             name = "proteins_centered")
stds <- rowSds(assay(scp, "proteins_centered"))
scp <- sweep(scp, MARGIN = 1, STATS = stds,
             FUN = "/", i = "proteins_centered",
             name = "proteins_scaled")
```

Different workflows = different results





One lab = one workflow



Manuscript in progress

Vanderaa Christophe, and Laurent Gatto. 2021. "Replication of Single-Cell Proteomics Data Reveals Important Computational Challenges." Expert Review of Proteomics, October, 1–9.

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